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ZUBA TOKEN

AUDIT

SECURITY ASSESSMENT

20TH December 2025

For



Making Blockchain, Defi And Web3 A Safer Place.



**Smart
Check**



SLITHER



**TRAIL
OF
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INTRODUCTION

Auditing Firm	 VITAL BLOCK SECURITY
Client Firm	 ZUBA THE SUPER DOG
Methodology	Automated Analysis, Manual Code Review
Language	Solidity
Contract Address	0xE1bf97baCF682AE17cA3E239aB68a7b525f994A4
Source Code Light	Verified
Centralization	Active ownership
Compiler Version	v0.8.30+commit.73712a01
Blockchain	 BINANCE CHAIN
Website	https://zubatoken.com
Twitter	https://x.com/zubatoken
Telegram	https://t.me/zubacommunitychat
Docs	https://zubatoken-jgydxne.gamma.site/
Prelim Report Date	December 18 TH 2025
Final Report Date	December 20 TH 2025

 Verify the authenticity of this report on our GitHub Repo: <https://www.github.com/vital-block>



Document Properties

Client	ZUBA TOKEN
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Auditors	Akhmetshin Marat, James BK, Ben Partrick , C. John
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Approved by	Prince Mitchell
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Version Info

Version	Date	Author(s)	Description
1.0	December 20 TH , 2025	C. John	Final Release
1.0-AP	December 20 Th , 2025	C. John	Release Candidate

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In the following, we show the specific pull request and the commit hash value used in this audit.

- [ZUBA TOKEN](#) (ZU760987)
- [Address: 0xE1bf97ba...525f994A4 | BscScan](#) (XCHJ86954)

About Vital Block

Vital Block Security provides professional, thorough, fast, and easy-to-understand smart contract security audit. We do in-depth and penetrative static, manual, automated, and intelligent analysis of the smart contract. Some of our automated scans include tools like ConsenSys MythX, Mythril, Slither, Surya. We can audit custom smart contracts, DApps, NFTs, etc (including the service of smart contract auditing). We are reachable at Telegram (<https://t.me/vitalblock>), Twitter (http://twitter.com/Vb_Audit), or Email (info@vitalblock.org).

Table 1.2: Vulnerability Severity Classification

Impact	High	Critical	High	Medium
	Medium	High	Medium	Low
	Low	Medium	Low	Low
		High	Medium	Low
		Likelihood		

Methodology

To standardize the evaluation, we define the following terminology based on the OWASP Risk Rating Methodology.

- Likelihood represents how likely a particular vulnerability is to be uncovered and exploited in the wild;
- Impact measures the technical loss and business damage of a successful attack;
- Severity demonstrates the overall criticality of the risk.

SCOPE OF WORK

Vital Block was consulted by **ZUBATOKEN** to conduct the smart contract audit of its **SOLIDITY (.SOL)** source code. The audit scope of work is strictly limited to the mentioned .Sol file only:

O.ZubaTheSuperDog.sol

 **External contracts and/or interfaces dependencies are not checked due to being out of scope.**

Verify audited contract's contract address and deployed link below:

Public Contract Address	
0xE1bf97baCF682AE17cA3E239aB68a7b525f994A4	
Contract Name	ZUBA THE SUPER DOG
Ticker	\$ZUBA
Total Supply	500,000,000,000

Executive Summary

The ZubaTheSuperDog contract implements a feature-rich ERC20 token with:

- * Fixed supply ($500B \times 1e18$) via ERC20Capped
- * Blacklist/whitelist controls
- * Buy/sell taxes (sent to configurable addresses)
- * LP pair detection for tax logic
- * Anti-bot protections (transaction delay, max holding, max tx size)
- * Owner-controlled toggles (trading enabled, bot protection)

While the design is generally sound and follows modern patterns (e.g., custom errors, modifiers), critical and high-risk issues were identified — particularly around tax redirection, inconsistent exemption logic, and reentrancy-adjacent hazards in `_beforeTokenTransfer`.



AUDIT METHODOLOGY

Smart contract audits are conducted using a set of standards and procedures. Mutual collaboration is essential to performing an effective smart contract audit. Here's a brief overview of Vital Block

Security auditing process and methodology:

CONNECT

- The onboarding team gathers source codes, and specifications to make sure we understand the size, and scope of the smart contract audit.

AUDIT

- Automated analysis is performed to identify common contract vulnerabilities. We may use the following third-party frameworks and dependencies to perform the automated analysis:
 - Remix IDE Developer Tool
 - Open Zeppelin Code Analyzer
 - SWC Vulnerabilities Registry
 - DEX Dependencies, e.g., Pancakeswap, Uniswap
- Simulations are performed to identify centralized exploits causing contract and/or trade locks.
- A manual line-by-line analysis is performed to identify contract issues and centralized privileges.

We may inspect below mentioned common contract vulnerabilities, and centralized exploits:

Centralized Exploits	○ Token Supply Manipulation ○ Access Control and Authorization ○ Assets Manipulation ○ Ownership Control ○ Liquidity Access ○ Stop and Pause Trading ○ Ownable Library Verification
------------------------------------	--

Common Contract Vulnerabilities

- Integer Overflow
- Lack of Arbitrary limits
- Incorrect Inheritance Order
- Typographical Errors
- Requirement Violation
- Gas Optimization
- Coding Style Violations
- Re-entrancy
- Third-Party Dependencies
- Potential Sandwich Attacks
- Irrelevant Codes
- Divide before multiply
- Conformance to Solidity Naming Guides
- Compiler Specific Warnings
- Language Specific Warnings

REPORT

- The auditing team provides a preliminary report specifying all the checks which have been performed and the findings thereof.
- The client's development team reviews the report and makes amendments to the codes.
- The auditing team provides the final comprehensive report with open and unresolved issues.

PUBLISH

- The client may use the audit report internally or disclose it publicly.

 It is important to note that there is no pass or fail in the audit, it is recommended to view the audit as an unbiased assessment of the safety of solidity codes.

Table 1.0 The Full Audit Checklist

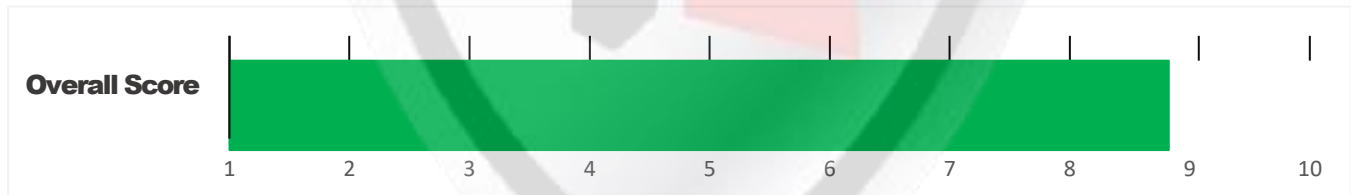
Category	Checklist Items
Basic Coding Bugs	Constructor Mismatch
	Ownership Takeover
	Redundant Fallback Function
	Overflows & Underflows
	Reentrancy
	Money-Giving Bug
	Blackhole
	Unauthorized Self-Destruct
	Revert DoS
	Unchecked External Call
	Gasless Send
	Send Instead Of Transfer
	Costly Loop
	(Unsafe) Use Of Untrusted Libraries
	(Unsafe) Use Of Predictable Variables
	Transaction Ordering Dependence
	Deprecated Uses
Semantic Consistency Checks	Semantic Consistency Checks
Advanced DeFi Scrutiny	Business Logics Review
	Functionality Checks
	Authentication Management
	Access Control & Authorization
	Oracle Security
	Digital Asset Escrow
	Kill-Switch Mechanism
	Operation Trails & Event Generation
	ERC20 Idiosyncrasies Handling
	Frontend-Contract Integration
	Deployment Consistency
	Holistic Risk Management
Additional Recommendations	Avoiding Use of Variadic Byte Array
	Using Fixed Compiler Version
	Making Visibility Level Explicit
	Making Type Inference Explicit
	Adhering To Function Declaration Strictly
	Following Other Best Practices

EXECUTIVE SUMMARY

Vital Block Security has performed the automated and manual analysis of the **ZUBA.Sol** code. The code was reviewed for common contract vulnerabilities and centralized exploits. Here's a quick audit summary:

Status	Critical ! 	Major " 	Medium # 	Minor \$ 	Unknown % 
Open	1	0	0	0	0
Acknowledged	0	2	2	1	0
Resolved	0	0	1	0	0
Noteworthy OnlyOwner Privileges	Set Taxes and Ratios, Airdrop, Set Protection Settings, Set Reward Properties, Set Reflector Settings, Set Swap Settings, Set Pair and Router				

ZUBA TOKEN Smart contract has achieved the following score: **87.0**



i Please note that smart contracts deployed on blockchains aren't resistant to exploits, vulnerabilities and/or hacks. Blockchain and cryptography assets utilize new and emerging technologies. These technologies present a high level of ongoing risks. For a detailed understanding of risk severity, source code vulnerability, and audit limitations, kindly review the audit report thoroughly.

i Please note that centralization privileges regardless of their inherited risk status - constitute an elevated impact on smart contract safety and security.

RISK CATEGORIES

Smart contracts are generally designed to hold, approve, and transfer tokens. This makes them very tempting attack targets. A successful external attack may allow the external attacker to directly exploit. A successful centralization-related exploit may allow the privileged role to directly exploit. All risks which are identified in the audit report are categorized here for the reader to review:

Risk Type	Definition
Critical 🚫	These risks could be exploited easily and can lead to asset loss, data loss, asset, or data manipulation. They should be fixed right away.
Major 🟡	These risks are hard to exploit but very important to fix, they carry an elevated risk of smart contract manipulation, which can lead to high-risk severity.
Medium 🟠	These risks should be fixed, as they carry an inherent risk of future exploits, and hacks which may or may not impact the smart contract execution. Low-risk re-entrancy-related vulnerabilities should be fixed to deter exploits.
Minor 🟢	These risks do not pose a considerable risk to the contract or those who interact with it. They are code-style violations and deviations from standard practices. They should be highlighted and fixed nonetheless.
Unknown 🟤	These risks pose uncertain severity to the contract or those who interact with it. They should be fixed immediately to mitigate the risk uncertainty.

All statuses which are identified in the audit report are categorized here for the reader to review:

Status Type	Definition
Open	Risks are open.
Acknowledged	Risks are acknowledged, but not fixed.
Resolved	Risks are acknowledged and fixed.



CENTRALIZED PRIVILEGES

Centralization risk is the most common cause of cryptography asset loss. When a smart contract has a privileged role, the risk related to centralization is elevated.

There are some well-intended reasons have privileged roles, such as:

- Privileged roles can be granted the power to `pause()` the contract in case of an external attack.
- Privileged roles can use functions like, `include()`, and `exclude()` to add or remove wallets from fees, swap checks, and transaction limits. This is useful to run a presale and to list on an exchange.

Authorizing privileged roles to externally-owned-account (EOA) is dangerous. Lately, centralization-related losses are increasing in frequency and magnitude.

- The client can lower centralization-related risks by implementing below mentioned practices:
- Privileged role's private key must be carefully secured to avoid any potential hack.
- Privileged role should be shared by multi-signature (multi-sig) wallets.
- Authorized privilege can be locked in a contract, user voting, or community DAO can be introduced to unlock the privilege.
- Renouncing the contract ownership, and privileged roles.
- Remove functions with elevated centralization risk.

 Understand the project's initial asset distribution. Assets in the liquidity pair should be locked.

Assets outside the liquidity pair should be locked with a release schedule.



AUTOMATED ANALYSIS

Symbol	Definition
	Function modifies state
	Function is payable
	Function is internal
	Function is private
	Function is important

```

**ZUBA THESUPERDOG** | Interface | |||
| L | totalSupply | External ! | |NO!|
| L | decimals | External ! | |NO!|
| L | symbol | External ! | |NO!|
| L | name | External ! | |NO!|
| L | getOwner | External ! | |NO!|
| L | balanceOf | External ! | ! |NO!|
| L | transfer | External ! | " !  |NO!|
| L | allowance | External ! | " ! |NO!|
| L | approve | External ! | " !  |NO!|
| L | transferFrom | External ! | " |NO!|
|||||
**IFactoryV2** | Interface | |||
| L | getPair | External ! | |NO!|
| L | createPair | External ! | " |NO!|
|||||
**IV2Pair** | Interface | |||
| L | factory | External ! | |NO!|
| L | getReserves | External ! | |NO!|
| L | sync | External ! | " |NO!|

```

|||||

| ****IRouter01**** | Interface | |||

| L | factory | External ! | |NO!|

| L | BNB | External ! | |NO!|

| L | addLiquidityBNB | External ! | # |NO!|

| L | addLiquidity | External ! | " |NO!|

| L | swapExactBNBTokens | External ! | # |NO!|

| L | getAmountsOut | External ! | |NO!|

| L | getAmountsIn | External ! | |NO!|

|||||

| ****IRouter02**** | Interface | IRouter01 |||

| L | swapExactTokensForBNBSupportingFeeOnTransferTokens | External ! | " |NO!|

| L | swapExactBNBForTokensSupportingFeeOnTransferTokens | External ! | # |NO!|

| L | swapExactTokensForTokensSupportingFeeOnTransferTokens | External ! | " ! |NO!|

| L | swapExactTokensForTokens | External ! | " |NO!|

|||||

| ****Protections**** | Interface | |||

| L | checkUser | External ! | " ! |NO!|

| L | setLaunch | External ! | " |NO!|

| L | setLpPair | External ! | " |NO!|

| L | ZUBA | External ! | " ! |NO!|

| L | removeSniper | External ! | " ! |NO!|

|||||

| ****Cashier**** | Interface | |||

| L | setRewardsProperties | External ! | " |NO!|

| L | tally | External ! | " ! |NO!|

| L | load | External ! | " ! |NO!|

| L | cashout | External ! | " ! |NO!|

| L | giveMeWelfarePlease | External ! | " ! |NO!|

| L | getTotalDistributed | External ! | " ! |NO!|

| L | getUserInfo | External ! | " ! |NO!|

| L | getUserRealizedRewards | External ! | " ! |NO!|

```

| L | getPendingRewards | External ! | ! | NO ! | |
| L | initialize | External ! | " ! | NO ! |
| L | getCurrentReward | External ! | | NO ! |
|||||
| **BNB** | Implementation | SafeMath |||
| L | <Constructor> | Public ! | ! | #S3 | NO ! |
| L | transferOwner | External ! | " ! | onlyOwner |
| L | renounceOwnership | External ! | " ! | NO ! |
| L | setOperator | Public ! | " ! | NO ! |
| L | renounceOriginalDeployer | External ! | " | NO ! |
| L | <Receive WBNB> | External ! | ! | #S3 | NO ! |
| L | totalSupply | External ! | ! | NO ! |
| L | decimals | External ! | ! | NO ! |
| L | symbol | External ! | ! | NO ! |
| L | name | External ! | ! | NO ! |
| L | getOwner | External ! | ! | NO ! |
| L | balanceOf | Public ! | ! | NO ! |
| L | allowance | External ! | ! | NO ! |
| L | approve | External ! | " ! | NO ! |
| L | _approve | Internal $ | " ! | |
| L | approveContractContingency | Public ! | " ! | onlyOwner |
| L | transfer | External ! | " ! | NO ! |
| L | transferFrom | External ! | " ! | NO ! |
| L | setNewRouter | External ! | " ! | onlyOwner |
| L | setLpPair | External ! | " ! | onlyOwner |
| L | setInitializers | External ! | " ! | onlyOwner |
| L | isExcludedFromFees | External ! | ! | NO ! |
| L | isExcludedFromDividends | External ! | ! | NO ! |
| L | isExcludedFromProtection | External ! | ! | NO ! |
| L | setDividendExcluded | Public ! | " ! | onlyOwner |
| L | setExcludedFromFees | Public ! | " ! | onlyOwner |

```

ZUBA TOKEN - 01 POSSIBLE OVERFLOW

Category	Severity ●	Location	Status
Status Mathematical Operations	Critical Findings	ZubaTheSuperDog()	Acknowledged

Description

C-1: Tax Tokens Sent to Uninitialized s_sellTaxLpAddress by Default → Permanent Loss

The constructor sets `s_buyTaxAddress`, but `s_sellTaxLpAddress` is never initialized, defaulting to `address(0)`. In `_beforeTokenTransfer`, sell taxes are unconditionally transferred to `s_sellTaxLpAddress`: **Impact**: Failed liquidity addition, loss of expected LP generation, possible stuck state.

```
super._transfer(from, s_sellTaxLpAddress, taxAmount);
```

If `s_sellTaxLpAddress == address(0)`, these tokens are **burned permanently** (sent to zero address). Since users cannot recover them, and the owner must manually call `setSellTaxLpAddress()` post-deployment, **early sellers will lose tax amounts irreversibly**.

This is especially dangerous because:

- * Owner may forget to set it before enabling trading.
- * No validation exists during tax transfer.
- * Zero address transfers in `_transfer` silently succeed (OpenZeppelin allows this).

Recommendation

- ✓ Fix 1 (Immediate): Initialize `s_sellTaxLpAddress` in constructor

```
constructor(string memory _name, string memory _symbol, address _buyTaxAddress)
    ERC20(_name, _symbol)
    ERC20Capped(MAX_SUPPLY)
    Ownable(msg.sender)
{
    if (_buyTaxAddress == address(0)) revert ZubaTheSuperDog__InvalidAddress();
    s_buyTaxAddress = _buyTaxAddress;
    s_sellTaxLpAddress = _buyTaxAddress; // or require explicit arg
    _mint(msg.sender, MAX_SUPPLY);
}
```

- ✓ Fix 2 (Defense-in-depth): Add runtime check

In `_beforeTokenTransfer`, before `_transfer`:

```
if (s_sellTaxLpAddress == address(0)) {
    revert ZubaTheSuperDog__InvalidAddress();
}
```

But better: make it a constructor argument to force explicit setup.

⚠ Also consider emitting an event when tax addresses change for transparency.

ZUBA TOKEN - 02 POSSIBLE OVERFLOW

Category	Severity ●	Location	Status
Inconsistency	Informational	ZubaTheSuperDog()	Acknowledged

_beforeTokenTransfer Modifies State Before Validation — Reentrancy-Adjacent Risk

Description

In _beforeTokenTransfer, tax transfers (super._transfer(...)) are executed before all validations complete

```
netAmount = value;

// Trading disabled check ↓ runs BEFORE tax transfer? Actually NO – tax calc happens AFTER.
// ...
if(s_lpPairs[to]) {
    // ... exemption check
    uint256 taxAmount = ...;
    netAmount -= taxAmount;
    super._transfer(from, s_sellTaxLpAddress, taxAmount); // ← STATE CHANGE HERE
}
// ... later: no further validations
```

While full reentrancy is unlikely (no external calls after), if s_sellTaxLpAddress is a malicious contract, it could:

Call back into ZubaTheSuperDog.transfer() before netAmount is fully computed or _transfer completes.

Exploit inconsistent state (e.g., balance updates delayed until after _beforeTokenTransfer).

Though OZ's _transfer is generally safe, performing external state changes early violates the Checks-Effects-Interactions pattern.

Impact

● High — Potential for reentrancy or state corruption if tax recipient is malicious or buggy.

Most severe if s_sellTaxLpAddress is upgradable or controlled by third party.

Recommendation

✅ Defer state-changing transfers until after all validations.

Recompute netAmount, but only record tax amounts in memory; apply transfers in _update or after _beforeTokenTransfer returns.

```
function _beforeTokenTransfer(address from, address to, uint256 value)
    internal
    returns (uint256 netAmount, address taxRecipient, uint256 taxAmount)
{
    netAmount = value;
    taxRecipient = address(0);
    taxAmount = 0;

    // ... [trading disabled check]

    if (s_lpPairs[to]) {
        if (!isExempt(from)) {
            taxAmount = (value * s_sellTaxPercentage) / 1000;
            if (taxAmount == 0 && value > 0) taxAmount = 1;
            netAmount -= taxAmount;
            taxRecipient = s_sellTaxLpAddress;
        }
    } else if (s_lpPairs[from]) {
        if (!isExempt(to)) {
            taxAmount = (value * s_buyTaxPercentage) / 1000;
            if (taxAmount == 0 && value > 0) taxAmount = 1;
            netAmount -= taxAmount;
            taxRecipient = s_buyTaxAddress;
        }
    }

    return (netAmount, taxRecipient, taxAmount);
}
```

ZUBA TOKEN - 03 POSSIBLE OVERFLOW

Category	Severity ●	Location	Status
Inconsistency	Medium	ZubaTheSuperDog()	Acknowledged

[setMaxHoldingPercentage](#) Allows 0% → Locks All Non-Exempt Transfers

Description

[setMaxHoldingPercentage\(uint16 newPercentage\)](#) only checks ≤ 100 , but allows 0.
If set to 0, then:

```
if (newBalance > ((0 * MAX_SUPPLY) / 100)) → if (newBalance > 0)
```

→ Any non-exempt transfer to a non-zero-balance recipient will revert with
[ZubaTheSuperDog__MaxHoldingExceeded\(\)](#).

This effectively **freezes all non-exempt activity**, worse than pausing.

Impact

● Medium — Denial of Service risk if owner misconfigures.

Accidental setting to 0 locks token; recovery requires owner to raise limit.

Recommendation

✓ Add lower bound:

```
require(newPercentage >= 1 && newPercentage <= 100, "Invalid percentage");
```

Or at minimum, emit warning event and add NatSpec: @dev Must be ≥ 1 .

ZUBA TOKEN - 04 POSSIBLE OVERFLOW

Category	Severity ●	Location	Status
Inconsistency	Medium	ZubaTheSuperDog()	Acknowledged

Low / QA Findings

Redundant `checkBlacklisted(to)` in `transfer/transferFrom`

Description

`checkBlacklisted` modifier is applied to both `msg.sender` and `to`, i.e.

```
checkBlacklisted(msg.sender)
checkBlacklisted(to)
```

But typically, **only the sender needs to be checked** — a blacklisted recipient can still receive tokens (e.g., for burning, migration). Preventing receipt may be overreach.

● Low — Overly restrictive policy, may confuse users.

Recommendation

✓ Consider removing `checkBlacklisted(to)` unless explicit design requires blocking receipt. If needed, document rationale clearly.

Gas Optimization Opportunities

G-1: Cache `balanceOf(recipient)` in `checkMaxHolding`

Current:

```
uint256 recipientBalance = balanceOf(recipient);
```

But `balanceOf` is a public view function → extra `load` under the hood.

✓ Replace with:

```
uint256 recipientBalance = _balances[recipient];
```

OPTIMIZATIONS | \$ZUBA

ID	Title	Category	Status
ERR	Logarithm Refinement Optimization	Gas Optimization	Acknowledged 
YUU	Checks Can Be Performed Earlier	Gas Optimization	Acknowledged 
BGH	Unnecessary Use Of SafeMath	Gas Optimization	Acknowledged 
JUP	Struct Optimization	Gas Optimization	Acknowledged 
WEE	Unused State Variable	Gas Optimization	Acknowledged 

OPTIMIZATION	ESTIMATED GAS SAVED	FREQUENCY	PRIORITY
Cache storage vars in <code>_update()</code>	800–1200	Every transfer	 High
Skip pair revalidation for non-DEX transfers	~100	Every transfer	 Medium
Optimize signer removal (if frequent)	~5,000	Rare	 Low
Combine <code>require()</code> → custom errors	~50–100	Admin ops	 Low

General Detectors

! Missing Zero Address Validation

Some functions in this contract may not appropriately check for zero addresses being used.



Attention
Required

! Consistent Solidity Version

This contract uses a conventional or very New version of Sol dependency



Attention
Required

- ✓ No compiler version inconsistencies found
- ✓ No unchecked call responses found
- ✓ No vulnerable self-destruct functions found
- ✓ No assertion vulnerabilities found
- ✓ No old solidity code found
- ✓ No external delegated calls found
- ✓ No external call dependency found
- ✓ No vulnerable authentication calls found
- ✓ No invalid character typos found
- ✓ No RTL characters found
- ✓ No dead code found
- ✓ No risky data allocation found
- ✓ No uninitialized state variables found
- ✓ No uninitialized storage variables found
- ✓ No vulnerable initialization functions found
- ✓ No risky data handling found
- ✓ No number accuracy bug found
- ✓ No out-of-range number vulnerability found
- ✓ No map data deletion vulnerabilities found
- ✓ No tautologies or contradictions found
- ✓ No faulty true/false values found
- ✓ No innacurate divisions found
- ✓ No redundant constructor calls found
- ✓ No vulnerable transfers found
- ✓ No vulnerable return values found
- ✓ No uninitialized local variables found
- ✓ No default function responses found
- ✓ No missing arithmetic events found
- ✓ No missing access control events found
- ✓ No redundant true/false comparisons found
- ✓ No state variables vulnerable through function calls found
- ✓ No buggy low-level calls found
- ✓ No expensive loops found
- ✓ No bad numeric notation practices found
- ✓ No missing constant declarations found
- ✓ No missing external function declarations found
- ✓ No vulnerable payable functions found
- ✓ No vulnerable message values found

Vulnerability Scan

REENTRANCY

✓ No reentrancy risk found

Severity

Minor

Confidence Parameter

Certain

✓ **NOT-RENOUNCED**: No additional amount of ZUBA TOKEN can be minted by a private wallet or contract.
(Which is normal for major contract utility options)

```

modifier checkTransactionDelay() {
    if (s_botProtection) {
        address sender = msg.sender;
        bool isExempt = whitelists[sender] ||
            s_lpPairs[sender] ||
            sender == owner() ||
            sender == s_buyTaxAddress;

        if (!isExempt) {
            uint256 lastTxTime = s_lastTransactionTime[sender];
            if (lastTxTime != 0 && lastTxTime + TRANSACTION_DELAY >= block.timestamp) {
                revert ZubaTheSuperDog__TransactionDelayNotPassed();
            }
            s_lastTransactionTime[sender] = block.timestamp;
        }
    }
}

```

Vulnerability Description

Scanning Line:

Identifier	Definition	Severity
CEN-02	Initial asset distribution	Minor \$ 

```
function _beforeTokenTransfer(address from, address to, uint256 value)
    internal returns (uint256 netAmount)
{
    netAmount = value;
    // If trading is disabled, only allow owner and whitelisted addresses to transfer
    if (!s_tradingEnabled) {
        bool isTradingEnabledExempt = from == owner() ||
            to == owner() ||
            whitelists[from] ||
            whitelists[to] ||
            from == address(this) ||
            to == address(this) ||
            from == s_buyTaxAddress ||
            to == s_buyTaxAddress;
        if (!isTradingEnabledExempt) {
            revert ZubaTheSuperDog__TradingDisabled();
        }
    }
}
```

Alleviation:

The Distribution asset was acknowledged and ultimately discarded by the **ZUBA TOKEN** team due to Earning severity. We consider the exhibit fully attended to as it doesn't impose any meaningful security concerns.

RECOMMENDATION

Clarify intent. If anti-bot, consider using time-based or unique buyer count instead.



Contract Creator Address:

0x5ee9926564cB98ab3EBc256bCadDD1372cb7f53f

Audited Files

ZUBA.Sol



Contracts Creator Hash:

TXN HASH

0x5654b857aa0622281d8d82a49688561982b3b1dc2f8b9df8e444db5041b406e9

Contracts:

Contract Address

ZUBA: 0xE1bf97baCF682AE17cA3E239aB68a7b525f994A4

MANUAL REVIEW

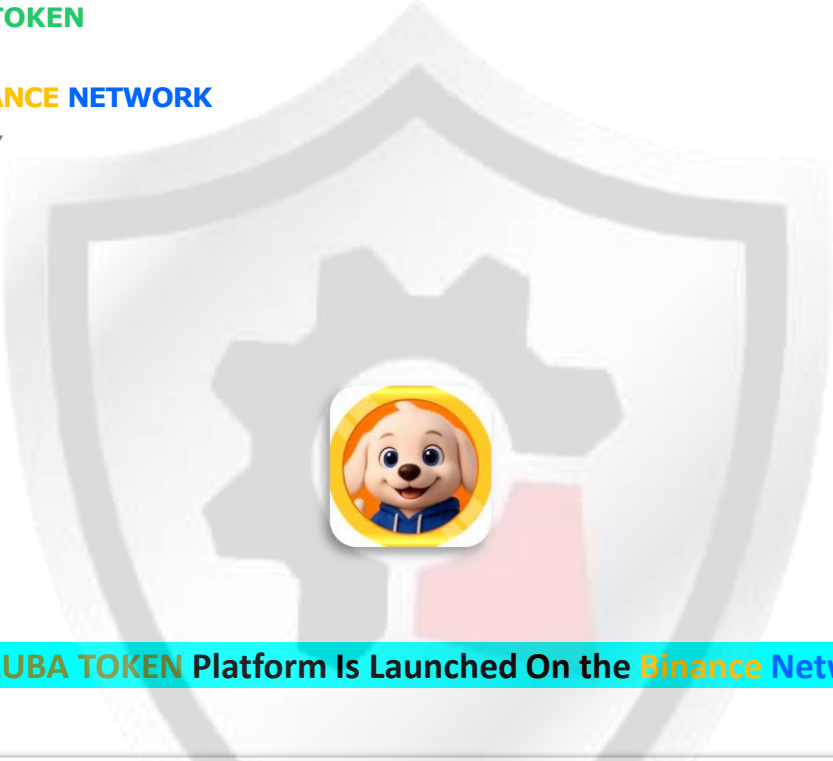
ZUBA TOKEN: ZUBA represents a new chapter in Africa's creative story, a bridge between culture, technology, and imagination. Our mission is to build a world where African anime and gaming stand proudly on the global stage, powered by community and innovation.

TOKEN NAME: ZUBA TOKEN

Ticker: \$ZUBA

Chain/Standard: BINANCE NETWORK

LAUNGE: SOLIDITY



The ZUBA TOKEN Platform Is Launched On the Binance Network





ISSUES CHECKING STATUS

Issue Description

Checking Status

1.	Compiler errors	PASSED
2.	Race Conditions and reentrancy. Cross-Function Race Conditions.	PASSED
3.	Possible Delay In Data Delivery.	PASSED
4.	Oracle calls	PASSED
5.	Front Running.	PASSED
6.	SOL Dependency.	PASSED
7.	Integer Overflow And Underflow.	PASSED
8.	DoS with Revert.	PASSED
9.	Dos With Block Gas Limit.	PASSED
10.	Methods execution permissions.	PASSED
11.	Economy Model of the contract.	PASSED
12.	The Impact Of Exchange Rate On the Move Logic.	PASSED
13.	Private use data leaks	PASSED
14.	Malicious Event log.	PASSED
15.	Scoping and Declarations	PASSED
16.	Uninitialized storage pointers	PASSED
17.	Arithmetic accuracy.	PASSED
18.	Design Logic.	PASSED
19.	Cross-Function race Conditions	PASSED
20.	Save Upon Move contract Implementation and Usage.	PASSED
21.	Fallback Function Security	PASSED



AUDIT RESULT

PASSED

SMART CONTRACT AUDIT OF ZUBA TOKEN

Identifier	Definition	Severity
CEN-02	Initial asset distribution	Minor 

All of the initially minted assets are sent to the contract deployer when deploying the contract. This can be an issue as the deployer and/or contract owner can distribute tokens without consulting the community.

```
function _update(address from, address to, uint256 value)
    internal
    override(ERC20, ERC20Capped)
{
    // When minting (from == address(0)), check cap
    if (from == address(0)) {
        if (totalSupply() + value > MAX_SUPPLY) revert ZubaTheSuperDog__ExceedMaxSupply();
    }
    super._update(from, to, value);
}
```

RECOMMENDATION

Project stakeholders should be consulted during the initial asset distribution process.

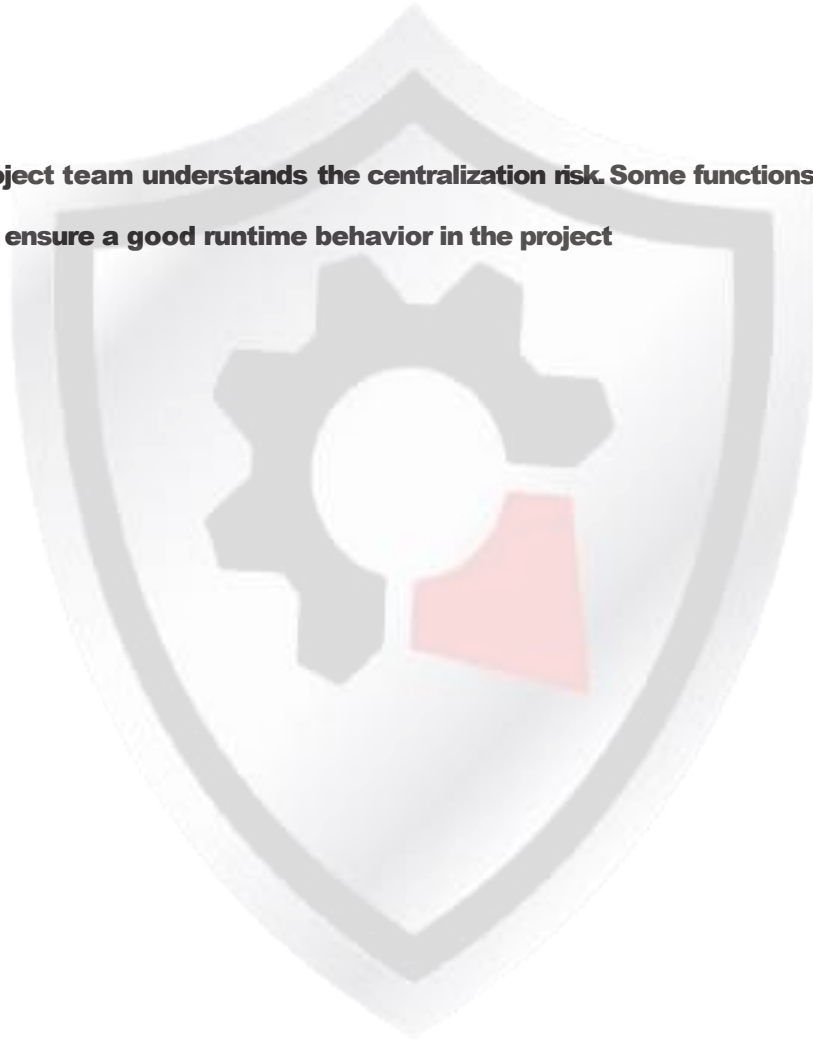
RECOMMENDATION

Deployer and/or contract owner private keys are secured carefully.

Please refer to PAGE-09 CENTRALIZED PRIVILEGES for a detailed understanding.

ALLEVIATION

The ZUBA TOKEN project team understands the centralization risk. Some functions are provided privileged access to ensure a good runtime behavior in the project



CERTIFICATE BY VITAL BLOCK SECURITY



CERTIFICATE OF COMPLIANCE

This certificate is presented to

ZUBA THESUPERDOG

This Project Contract Code Has Been Verified

This Safety Certificate Is Only Valid For >

0XE1BF97BACF682AE17CA3E239AB68A7B525F994A4

MAXIMUM SCORE ACHIEVED

SCORE
87



Identifier	Definition	Severity
COD-10	Third Party Dependencies	Minor 

Smart contract is interacting with third party protocols e.g., Pancakeswap router, cashier contract, protections contract. The scope of the audit treats third party entities as black boxes and assumes their functional correctness. However, in the real world, third parties can be compromised, and exploited. Moreover, upgrades in third parties can create severe impacts, e.g., increased transactional fees, deprecation of previous routers, etc.

RECOMMENDATION

Inspect and validate third party dependencies regularly, and mitigate severe impacts whenever necessary.



DISCLAIMERS

Vital Block provides the easy-to-understand audit of Solidity, Move and Raw source codes (commonly known as smart contracts).

The smart contract for this particular audit was analyzed for common contract vulnerabilities, and centralization exploits. This audit report makes no statements or warranties on the security of the code. This audit report does not provide any warranty or guarantee regarding the absolute bug-free nature of the smart contract analyzed, nor do they provide any indication of the client's business, business model or legal compliance. This audit report does not extend to the compiler layer, any other areas beyond the programming language, or other programming aspects that could present security risks. Cryptographic tokens are emergent technologies, they carry high levels of technical risks and uncertainty. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis. This audit report could include false positives, false negatives, and other unpredictable results.

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ABOUT VITAL BLOCK

Vital Block provides intelligent blockchain Security Solutions. We provide solidity and Raw Code Review, testing, and auditing services. We have Partnered with 15+ Crypto Launchpads, audited 50+ smart contracts, and analyzed 200,000+ code lines. We have worked on major public blockchains e.g., Ethereum, Binance, Cronos, Doge, Polygon, Avalanche, Metis, Fantom, Bitcoin Cash, Aptos, Oasis, etc.

Vital Block is Dedicated to Making Defi & Web3 A Safer Place. We are Powered by Security engineers, developers, UI experts, and blockchain enthusiasts. Our team currently consists of 5 core members, and 4+ casual contributors.

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